

Library Database System Overview

1. Business Context and Problem

Business Name and Website:

Business Name: State Library Victoria

Website: <https://www.slv.vic.gov.au/>

Business Context:

State Library Victoria is a publicly funded research library that provides free access to information and cultural resources. It serves students, researchers, and the general public, managing a large collection of physical and digital materials such as books, archives, and online databases. The organisation relies on information systems to manage resources, users, and service access.

Main Business:

The primary function of State Library Victoria is to manage and preserve information and cultural collections, while providing accessible services that support research, education, and public use.

Business Processes, Functions, and Activities:

The organisation manages library resources by acquiring, cataloguing, and maintaining physical and digital materials for efficient retrieval. Users search the catalogue using attributes such as title, author, and subject. Borrowing and return processes track loans, due dates, and item availability, while reservations allow users to place holds on unavailable items. The library also provides access to digital resources such as eBooks and databases, along with staff support and public programs such as events and exhibitions.

Scenarios and Use Cases:

Typical scenarios include users registering as members, searching for resources, borrowing and returning items, and placing reservations. Users may also access digital resources online, while staff maintain and update catalogue records. The library also supports participation in events and programs.

Business Problem:

State Library Victoria requires an efficient database system to manage large volumes of data related to resources, users, and services. The system must support accurate storage, organisation, and retrieval of information, while enabling processes such as membership management, borrowing, returns, and reservations.

2. Entities and Attributes

Initial Dataset

Member {member_id (PK), first_name, last_name, address, phone_numbers, email, membership_date, expiry_date, membership_status}

Staff {staff_id (PK), first_name, last_name, role, phone, email, employment_date}

ResourceCopy {resource_id (PK), copy_number (PK), title, resource_type, publication_year, language, subject_area, status, acquisition_date, copy_status, location_code, location_name}

Loan {loan_id (PK), member_id (FK), resource_id (FK), copy_number (FK), staff_id (FK), issue_date, due_date, return_date, loan_status}

Reservation {reservation_id (PK), member_id (FK), resource_id (FK), staff_id (FK), reservation_date, reservation_status}

Event {event_id (PK), staff_id (FK), event_name, event_date, event_time, location, description, event_capacity}

Initial Analysis

- Repeating group in Member through phone_numbers (violates 1NF)
- Partial dependency in ResourceCopy (violates 2NF)
- Transitive dependency in ResourceCopy, where location_name depends on location_code (violates 3NF)
- Many-to-many relationship between Member and Event, resolved later through EventRegistration

3. Normalisation

First Normal Form (1NF)

Removes the repeating group phone_numbers from Member by moving phone numbers into a separate entity. Staff already satisfies 1NF, as it stores a single contact phone number per staff member as an atomic attribute.

...

Member {member_id (PK), first_name, last_name, address, email, membership_date, expiry_date, membership_status}

MemberPhone {member_id (PK, FK), phone_number (PK)}

...

Second Normal Form (2NF)

Splits ResourceCopy into Resource and ResourceCopy to resolve the partial dependency, where attributes such as title, resource_type, publication_year, language, subject_area, and status depended only on resource_id rather than the full composite key.

...

Resource {resource_id (PK), title, resource_type, publication_year, language, subject_area, status}

ResourceCopy {resource_id (PK, FK), copy_number (PK), acquisition_date, copy_status, location_code, location_name}

...

Third Normal Form (3NF)

Splits ResourceCopy and Location to resolve the transitive dependency, where location_name depended on the non-key attribute location_code.

...

Location {location_code (PK), location_name}

ResourceCopy {resource_id (PK, FK), copy_number (PK), acquisition_date, copy_status, location_code (FK)}

...

Remaining Entities

The remaining entities (Staff, Loan, Reservation, Event, and EventRegistration) already meet 3NF and don't need any changes. Each one has atomic attributes and a clear primary key (1NF). Because the primary keys are single attributes, there are no partial dependencies (2NF). All non-key attributes depend directly on the primary key, with no transitive dependencies (3NF).

Resolved many-to-many relationship

The many-to-many relationship between Member and Event is resolved through EventRegistration.

...

EventRegistration {registration_id (PK), member_id (FK), event_id (FK), registration_date, attendance_status}

...

Normalized Schema

Member {member_id (PK), first_name, last_name, address, email, membership_date, expiry_date, membership_status}

MemberPhone {member_id (PK, FK), phone_number (PK)}

Staff {staff_id (PK), first_name, last_name, role, phone, email, employment_date}

Resource {resource_id (PK), title, resource_type, publication_year, language, subject_area, status}

Location {location_code (PK), location_name}

ResourceCopy {resource_id (PK, FK), copy_number (PK), acquisition_date, copy_status, location_code (FK)}

Loan {loan_id (PK), member_id (FK), resource_id (FK), copy_number (FK), staff_id (FK), issue_date, due_date, return_date, loan_status}

Reservation {reservation_id (PK), member_id (FK), resource_id (FK), staff_id (FK), reservation_date, reservation_status}

Event {event_id (PK), staff_id (FK), event_name, event_date, event_time, location, description, event_capacity}

EventRegistration {registration_id (PK), member_id (FK), event_id (FK), registration_date, attendance_status}

4) Entity Relationship Diagram (ERD)



